

Testing and Quality Assurance -- Guided Notes ANSWER KEY

Unit tp-4.3 | CompTIA Network+ N10-009 Obj. FC0-U71 4.3 | N10-008 FC0-U71 4.3

For instructor use / distribute after students complete the notes.

Question 1

QA (Quality Assurance) is the process of verifying that software meets its _____ before release. A bug (defect) found in _____ is far less expensive to fix than one found in _____. The earlier a defect is found in the SDLC, the _____ it costs to fix.

Answer: Requirements (specifications); testing; production; cheaper (less).

Question 2

_____ testing verifies each individual function or module works correctly in isolation. _____ testing verifies that components work correctly when combined. _____ testing verifies the complete system meets user requirements from end to end (also called acceptance testing).

Answer: Unit; integration; system (end-to-end).

Question 3

_____ testing is performed without knowledge of the internal code. The tester interacts with the software as a _____ would. _____ testing is performed with full knowledge of the source code. _____ (gray-box) testing is a combination of both approaches.

Answer: Black-box; user; white-box; gray-box.

Question 4

_____ testing verifies the same software functions correctly across different browsers, operating systems, and hardware. _____ testing subjects the system to heavy load to find failure points. _____ testing verifies performance under expected peak conditions to ensure response times remain acceptable.

Answer: Compatibility (cross-platform); stress (load); performance.

Question 5

A test case describes a specific _____, the steps to reproduce it, the _____ result, and the actual result. A bug report documents a defect including steps to _____, expected vs. actual behavior, and _____ level to help developers prioritize fixes.

Answer: Scenario; expected; reproduce; severity.

Question 6

_____ testing runs automatically every time code is committed and checks that new changes do not break existing features. _____ testing involves real users testing the software in their own environment before release to validate it meets their needs. These are also called alpha and _____ testing.

Answer: Regression; user acceptance testing (UAT); beta.

Question 7

Security testing attempts to find _____ in the software before attackers do. _____ testing simulates attacks to find exploitable vulnerabilities. _____ analysis reviews source code for security flaws without executing it. A recently published list of the top 10 web application vulnerabilities is maintained by _____.

Answer: Vulnerabilities; penetration (pen); static (SAST); OWASP.

Question 8 -- Real World Application

REAL WORLD: A development team releases a new feature in their mobile banking app without running regression tests because they assumed the new code only affected the new feature. After release, customers report they can no longer log in because authentication broke. What type of testing should have caught this and describe how it works.

Answer: Regression testing should have caught this before release. Regression testing re-runs the existing suite of automated test cases against the new code to verify that new changes have not broken any previously working functionality. In this case, a regression test suite covering the login/authentication flow would have automatically run after the feature code was merged. When the test run executed the login test case (enter valid credentials, expect dashboard), it would have received an unexpected error instead -- the test would fail and the CI/CD pipeline would have blocked deployment. The team would have been alerted before the code ever reached production that authentication had broken. The lesson: every code change, no matter how isolated it appears, must pass the regression suite before production deployment.